

Variable Dictionary

File: worker_panel_lagos.parquet | **Path:** Data/CBA_RAIS_firm_level/

Observations: 19,598,473 worker-years | **Unique workers (all years):** 5,723,160 |

Establishments: 16,472 (balanced) | **Years:** 2009–2016

All observations have *empem3112* = 1 (employed on December 31). This is a stock-of-workers panel. The 5,723,160 unique workers is a cumulative count across all years — many workers appear in multiple years.

Year	Obs	Unique workers	Establishments
2009	2,290,841	2,284,339	16,472
2010	2,454,406	2,448,653	16,472
2011	2,564,403	2,558,690	16,472
2012	2,565,124	2,558,835	16,472
2013	2,566,450	2,560,553	16,472
2014	2,529,443	2,523,797	16,472
2015	2,379,563	2,374,097	16,472
2016	2,248,243	2,243,227	16,472

Note: obs ≈ unique workers each year. The small gap (~5–6K workers, or ~0.2 %, appear at more than one establishment in the same year) is consistent with the one-spell-per-worker selection in 011_rais_to_firm.do allowing a small number of exceptions. The establishment count is identical every year — this is a **balanced panel on the firm side** (in_balanced_panel restriction).

Part 1 — Raw Variables

Variables taken directly from RAIS microdata with no transformation beyond stripping the leading '1' from 15-digit establishment IDs.

1.1 Identifiers

Variable	Type	Description
identificad	VARCHAR	14-digit establishment identifier (CNPJ-base from RAIS). Note: some raw RAIS files carry 15-digit IDs with a leading '1' — that prefix has been stripped.
PIS	VARCHAR	Worker identifier (PIS/PASEP social insurance number).

1.2 Wages

remdezembro and remmedia are in **minimum wages (salários mínimos)**. remdezr and remmedr are in **nominal BRL** — the 'r' suffix in RAIS stands for *remuneração* (remuneration), not real/deflated. salcontr and ultrem are also in **nominal BRL**.

Variable	Type	Min	Mean	Max	Description
remdezembro	DOUBLE	0	4.50 MW	150 MW	December earnings in minimum wages.

Variable	Type	Min	Mean	Max	Description
remmedia	DOUBLE	0	4.23 MW	150 MW	Average monthly earnings in minimum wages.
remdezzr	DOUBLE	0	2,902 BRL	132,007	December earnings in nominal BRL (raw RAIS field; 'r' = remuneração, not real/deflated).
remmedr	DOUBLE	0	2,736 BRL	130,875	Average monthly earnings in nominal BRL (same naming convention as remdezzr).
salcontr	DOUBLE	0	2,035 BRL	9,999,999	Contracted salary in nominal BRL.
ultrem	DOUBLE	0	—	4,081,467	Last remuneration received, nominal BRL.

1.3 Work Schedule

Variable	Type	Min	Mean	Max	Description
horascontr	TINYINT	1	41.4	44	Contracted weekly hours.

1.4 Contract Type

Variable	Type	Key Values	Description
tpvinculo	TINYINT	10=CLT private (93.2 %), 20=rural/CLT rural (3.0 %), 55=apprentice/Lei 10.097 (1.5 %), 60=fixed-term/Lei 9.601 (1.5 %), 70=rural other (0.3 %), 90=fixed-term other (0.2 %), 30/31/35=public statutory (0.1 %), 40/80/15/25=other (0.04 %)	Employment bond type (vínculo empregatício).
tipoadm	TINYINT	0=not admitted this year/transfer (77.2 %), 2=first job/new hire (17.0 %), 1=rehire (3.8 %), 4=transfer (1.9 %)	Admission type for workers hired during the year.
tiposal	TINYINT	1=monthly (77.1 %), 5=hourly (21.0 %), 6=daily (0.9 %), 4=task-based (0.4 %), 7=commission (0.4 %)	Salary payment type (tipo de salário).

1.5 Date Fields

Variable	Type	Format	Example	Description
dtadmissao	VARCHAR	DDMMYYYY	01052006	Date of admission to current establishment.
dtnascimento	VARCHAR	DDMMYYYY	31081966	Worker date of birth.

1.6 Worker Demographics

Variable	Type	Key Values	Description
genero	TINYINT	1=male (68.1 %), 0=female (31.9 %)	Gender (raw RAIS field).

Variable	Type	Key Values	Description
raca_cor	TINYINT	2=branca/white (58.4 %), 8=parda/mixed (31.3 %), 4=preta/black (5.4 %), 9=indígena (3.8 %), 6=amarela/Asian (0.9 %), 99=undeclared, -1=missing	Race/color (cor ou raça).
nacionalidad	TINYINT	10=Brazilian (99.7 %), 40/45/20/21/80=foreign	Nationality code.
idade	DOUBLE	0–100, mean 35.9	Age as reported in RAIS record.

1.7 Education

Variable	Type	Key Values	Description
grinstrucao	TINYINT	7=complete secondary (43.9 %), 9=complete higher (18.5 %), 5=complete primary (9.1 %), 4=incomplete secondary (7.7 %), 6=incomplete higher (7.5 %), 8=incomplete post-grad (4.5 %), 3=incomplete primary (4.0 %), 2=illiterate/no schooling (3.6 %)	Education level — raw RAIS grau de instrução code.

1.8 Disability

Note: portdefic flags ~25 % of workers (likely includes administrative filings and partial records). Use d_disabled (Part 2) for analysis.

Variable	Type	Key Values	Description
portdefic	TINYINT	0=no (74.5 %), 1=yes (25.5 %)	Raw RAIS disability carrier field (porte de deficiência). High share may reflect administrative coding.
tpdefic	TINYINT	0=none (97.8 %), 1=physical (1.1 %), 2=hearing (0.6 %), 6=intellectual (0.2 %), 3=visual (0.2 %), 4=rehabilitated (0.1 %), 5=multiple (0.03 %)	Disability type (tipo de deficiência).

1.9 Occupation

Variable	Type	Description
ocup2002	INTEGER	CBO 2002 occupation code at the 6-digit level (e.g., 784205=farm workers, 411005=general office clerks, 622110=crop farmers).

1.10 Establishment Characteristics

Variable	Type	Key Values	Description
clascnae20	VARCHAR	5-digit CNAE 2.0 code (e.g., 10716=poultry processing, 86101=hospitals, 29107=auto manufacturing)	Economic activity classification (CNAE 2.0).
municipio	INTEGER	IBGE 6-digit code; 2,096 distinct municipalities	Municipality of the establishment.

Variable	Type	Key Values	Description
tamestab	TINYINT	1=1–4 emp, 2=5–9, 3=10–19, 4=20–49, 5=50–99, 6=100–249, 7=250–499, 8=500–999, 9=1000+	Establishment size category.
natjuridica	SMALLINT	2062=Ltda (41.0 %), 2054=S.A. fechada (24.0 %), 2046=EIRELI (12.1 %), 3999=other private (6.6 %), 2011=S.A. aberta (4.5 %)	Legal nature of the establishment (natureza jurídica).

1.11 Employment Status & Separation

Near-universal `causadesli = 0` is consistent with the `empem3112 = 1` selection. The small number of non-zero values likely reflects administrative corrections or mid-year separations with same-year rehire.

Variable	Type	Key Values	Description
empem3112	TINYINT	1 for all obs	Employed on December 31 — the selection criterion defining the panel.
causadesli	TINYINT	0=still employed (99.6 %), 71=death (0.4 %), 80=retirement (0.02 %), 78=transfer (0.02 %)	Cause of separation (causa do desligamento).
mesdesli	TINYINT	All 0	Month of separation (all zero — consistent with Dec 31 employment).
diadesli	DOUBLE	All 0	Day of separation (all zero).

1.12 Leave / Absence Records

RAIS tracks up to three leave spells per worker-year. `-1` = not applicable to this spell index; 99 in `causafast` = employed but no leave during the year.

Variable	Type	Key Values	Description
causafast1	TINYINT	99=no leave (67.2 %), -1=N/A (21.2 %), 40=illness (8.9 %), 50=work accident (1.1 %), 10=maternity (0.7 %), 70=other (0.6 %)	Cause of first leave spell.
causafast2	TINYINT	Same codes	Cause of second leave spell.
causafast3	TINYINT	Same codes	Cause of third leave spell.
mesiniaf1	TINYINT	-1=N/A, 1–12=month	Start month of first leave spell.
mesiniaf2	TINYINT	-1=N/A, 1–12=month	Start month of second leave spell.
mesiniaf3	TINYINT	-1=N/A, 1–12=month	Start month of third leave spell.
mesfimaf1	TINYINT	-1=N/A, 1–12=month	End month of first leave spell.
mesfimaf2	TINYINT	-1=N/A, 1–12=month	End month of second leave spell.
mesfimaf3	TINYINT	-1=N/A, 1–12=month	End month of third leave spell.

Part 2 — Derived Variables

Variables constructed from raw RAIS fields during data cleaning (011_rais_to_firm.do / 011c_worker_panel.py).

2.1 Panel Identifier

Variable	Type	Values	Description
year	BIGINT	2009–2016	Calendar year of the observation. Added when stacking annual RAIS cross-sections into the panel.

2.2 Wages — Deflated & Log

remdezr and remmedr (raw RAIS, nominal BRL) are deflated to December 2015 BRL by dividing by the year-specific IPCA index (normalised to 1 in December 2015): $\text{lr_remdezr} = \log(\text{remdezr} / \text{IPCA_t})$. salcontr_m adjusts salcontr to a monthly basis using contracted hours, then is similarly deflated. Log versions are missing when the level is zero.

Variable	Type	Min	Mean	Max	Nulls	Description
salcontr_m	DOUBLE	0	—	75,968,082	—	Monthly contracted salary in nominal BRL (hours-adjusted from salcontr).
lr_remdezr	DOUBLE	5.37	7.79	11.73	797,129	$\log(\text{remdezr} / \text{IPCA_t})$ — log December earnings in Dec 2015 BRL.
lr_remmedr	DOUBLE	5.26	7.72	11.72	417,774	$\log(\text{remmedr} / \text{IPCA_t})$ — log average monthly earnings in Dec 2015 BRL.
lr_salcontr_m	DOUBLE	−4.61	7.68	18.49	7,043,827	$\log(\text{salcontr_m} / \text{IPCA_t})$ — log monthly contracted salary in Dec 2015 BRL. High nulls because salcontr is often zero.

2.3 Work Schedule

Variable	Type	Values	Description
hours_ge40	TINYINT	0/1 (84.9 % = 1)	1 if contracted hours $\geq$ 40/week. Derived from horascontr.

2.4 Contract Type Dummies

Binary indicators derived from tpvinculo. Reference category is CLT private (tpvinculo = 10).

Variable	Type	Share = 1	Source values (tpvinculo)	Description
d_apprentice	TINYINT	1.5 %	55	Apprentice contract (Lei 10.097).
d_fixed_term	TINYINT	1.8 %	60, 90, 50, 95, 96, 97	Fixed-term contract (contrato por prazo determinado).
d_rural	TINYINT	3.4 %	20, 70	Rural sector employment (Lei 5889/73).

Variable	Type	Share = 1	Source values (tpvinculo)	Description
d_public	TINYINT	0.1 %	30, 31, 35	Public sector statutory contract.
d_other_contract	TINYINT	0.05 %	40, 80, 15, 25	Residual/other contract type.

2.5 Tenure

Variable	Type	Range / Categories	Source	Description
tempempr	DOUBLE	1.1–599.9 months (mean 71.8)	dtadmissao	Tenure at the establishment in months.
tenure_bin	VARCHAR	lt1yr, 1_2yr, 3_5yr, 5_10yr, 10_20yr, 20plus	tempempr	Tenure binned into duration bands.

2.6 Worker Demographics

Variable	Type	Key Values	Source	Description
d_male	TINYINT	0/1 (68.1 % = 1)	genero	Binary gender indicator.
race_group	VARCHAR	branca (58.4 %), parda (31.3 %), preta (5.4 %), other_missing (5.0 %)	raca_cor	Race recoded; amarela, indígena, undeclared → 'other_missing'.
age	DOUBLE	0–109, mean 35.7	dtnascimento	Age computed from birth date and reference date.
age_bin	VARCHAR	14_24 (16.3 %), 25_34 (35.2 %), 35_44 (25.8 %), 45_54 (16.7 %), 55plus (5.9 %), missing (0.01 %)	age / idade	Age binned into five groups.

2.7 Education

Variable	Type	Key Values	Source	Description
educ_bin	VARCHAR	0_no_hs=less than HS (32.2 %), 1_hs=HS complete (48.4 %), 2_higher=higher education (19.4 %)	grinstrucao	Education recoded into three bins.

2.8 Disability

Variable	Type	Values	Source	Description
d_disabled	TINYINT	0/1 (2.2 % = 1, 435K workers)	tpdefic	Clean binary: 1 if tpdefic ≠ 0. Preferred over portdefic for analysis.

2.9 Occupation

Variable	Type	Description	Source
ocup4	INTEGER	CBO 2002 occupation at the 4-digit family level (first four digits of ocup2002).	ocup2002

Notes on Construction

- **Sample:** Lagos (2021) sample — establishments in the lagos_sample selection, restricted to workers employed on December 31 (empem3112 = 1).
- **Wage deflation:** lr_remdezr and lr_remmedr use Brazil's IPCA index, base = December 2015. The raw level variables remdezr and remmedr are in nominal BRL.
- **portdefic vs d_disabled:** portdefic has ~25 % flagged (likely includes administrative disability filings); d_disabled = (tpdefic ≠ 0) gives the standard 2.2 % rate.
- **Leave coding:** -1 in mesiniaf*/mesfimaf* means no leave spell of that index. 99 in causafast* means the worker was employed but had no leave.
- **One spell per worker per year:** n_obs ≈ n_unique_workers in every year. A small residual of ~5–6K workers per year (~0.2 %) appear at more than one establishment in the same year. This is consistent with the one-spell-per-worker selection algorithm in 011_rais_to_firm.do (ranked by hours, then hourly wage, then random tiebreaker with seed 12345), which selects one spell per worker-firm pair but does not enforce uniqueness across firms.
- **Balanced firm panel:** the establishment count is exactly 16,472 in every year (2009–2016), reflecting the in_balanced_panel restriction.